STAFF REVIEW · H446 284 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Transport layer.

2. An IP address for the name, when the appropriate DNS records exist.

3. Yes.

4. Capacity can be used by other traffic when a user is idle, improving efficiency.

I Do — source worked steps

1. Route — The network may forward them along different paths, so arrival order is not guaranteed.

2. Receive — TCP uses sequence information to buffer out-of-order data.

3. Deliver — When missing earlier bytes arrive, TCP makes data available to the application in the correct order: first 1, then 2.

We Do — guided CHECK

1. Recover
TCP can cause the missing data to be retransmitted; once the gap is filled, data is delivered in order.

You Do — source mark schemes

1. [2] Packet switching forwards separate packets using shared network links [1]; circuit switching establishes a dedicated or reserved connection/path resources for a session [1].

2. [4] TCP identifies the out-of-order data from sequence information [1], buffers later data while earlier bytes are missing [1], releases the ordered stream once the gap is filled [1], so the application receives 1,2,3 [1]. This uses simplified labelled segments; TCP sequence numbers actually identify byte positions.

3. [4] Packet switching allows links to be shared efficiently or can route around a failed path [2 with explanation]; agreed communication standards define consistent protocols/formats [1], allowing independently produced devices and software to interoperate [1].

Exit Ticket — source answer

1. False. IP is best effort; protocols such as TCP offer reliable ordered delivery.